

Md. Almabud **JUWAL**  
Senior Software Engineer | B.Sc. Computer Science and Engineering

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 Dhaka, Bangladesh

 EDUCATION

|      |                                                                                                                                                                                                                                                                                    |
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| 2020 | B.Sc. in Computer Science and Engineering                                                                                                                                                                                                                                          |
| 2016 | <div><div>➤ Daffodil International University, Dhaka, Bangladesh</div><div>➤ CGPA : 3.77 / 4.00</div><div>➤ Relevant coursework : Artificial Intelligence, Algorithms, Data Structures, Software Engineering, Compiler Design, Discrete Mathematics, Computer Networks</div></div> |

 ACADEMIC & RESEARCH INTERESTS

INTERESTS : Artificial Intelligence, Machine Learning, Data Science, Software Engineering, Distributed Systems

 SKILLS

|                              |                                                          |
|------------------------------|----------------------------------------------------------|
| Programming                  | Python, SQL, JavaScript/TypeScript, Go, C/C++            |
| Backend & Frameworks         | FastAPI, Django, Django REST Framework, Celery           |
| Data & Databases             | PostgreSQL, Pandas, Redis, MongoDB, Neo4j, Elasticsearch |
| Infrastructure & Engineering | Docker, RabbitMQ, Git, CI/CD, Jenkins, Ansible           |
| Frontend                     | Angular                                                  |

 EXPERIENCE

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| Present<br>Jun 2024  | <div>Senior Software Engineer   Cogwheel Analytics, Colorado, United States</div> <div><div>➤ Architected and led the development of the next-generation <b>Cogwheel Analytics</b> platform as a <b>modular monolith</b> following <b>Clean Architecture</b> principles, with clear separation of concerns and reusable handling for authentication, pagination, and request validation, reducing repetitive implementation work and accelerating feature delivery</div><div>➤ Lead the development and maintenance of the <b>Data Cleaning</b> service, implementing data-validation and transformation workflows to improve data integrity and consistency</div><div>➤ Design and implement backend services and APIs using Python, FastAPI, and PostgreSQL, with emphasis on maintainability, reliability, and performance</div><div>➤ Build and maintain asynchronous data-processing and web-scraping workflows using Pandas, Celery, and Scrapy</div></div> |
| May 2024<br>Jul 2023 | <div>Senior Software Engineer   Brainstation 23 Ltd, Dhaka, Bangladesh</div> <div><div>➤ Led the turnaround of a delayed project by mentoring <b>3 junior engineers</b>, restructuring task allocation, and redesigning the system architecture, enabling successful delivery of the second version within <b>6 months</b></div><div>➤ Designed software architectures for <b>Task Station</b> and <b>Sayvery</b>, with focus on scalability, maintainability, and clear separation of responsibilities</div><div>➤ Managed the technical backlog by translating product and stakeholder requirements into prioritized engineering tasks aligned with project goals and delivery timelines</div><div>➤ Designed and implemented web-scraping and invoice-parsing solutions to automate data collection and extraction</div></div>                                                                                                                                 |
| Jun 2023<br>Oct 2021 | <div>Software Engineer   Brainstation 23 Ltd, Dhaka, Bangladesh</div> <div><div>➤ Improved system performance by approximately <b>10%</b> through database-query optimization</div><div>➤ Improved API response performance by approximately <b>5%</b> by introducing application-level caching</div><div>➤ Upgraded Python and Django versions and refactored the existing codebase to improve maintainability and compatibility</div><div>➤ Contributed to real-time synchronization between backend and frontend systems using WebSockets and Go, while mentoring an Associate Software Engineer</div></div>                                                                                                                                                                                                                                                                                                                                                   |

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| Sep 2021<br>Jul 2020 | <b>Junior Software Engineer   Tizok Private Limited, Dhaka, Bangladesh</b> <ul style="list-style-type: none"> <li>➤ Reduced processing time for <b>3.3 million records</b> from approximately <b>29–35 minutes to 2–4 minutes</b> using PostgreSQL COPY and database triggers</li> <li>➤ Reduced database query response time by approximately <b>50%</b> through query optimization</li> <li>➤ Contributed to backend and database architecture, including the implementation of a custom authentication backend and graph-based data solutions using Neo4j</li> <li>➤ Automated deployment workflows using Jenkins and Ansible, reducing manual deployment effort and improving release consistency</li> </ul> |
| Apr 2020<br>Dec 2019 | <b>Web Developer   Wizdoor Inc, Dhaka, Bangladesh</b> <ul style="list-style-type: none"> <li>➤ Contributed to backend and database architecture while coordinating implementation with frontend and design teams</li> <li>➤ Developed a real-time chat application using Django Channels and WebSockets</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                               |

## PROJECTS

### COGWHEEL ANALYTICS

Jun 2024 – Present

<https://cogwheelanalytics.com>

Enterprise analytics platform for automated data ingestion, validation, processing, and reporting. The next-generation platform is designed as a modular monolith following Clean Architecture principles, with reusable application components and asynchronous processing to support maintainable and scalable analytics workflows.

### REHIVE

Oct 2021 – Feb 2023

<https://rehive.com/>

Contributed to a South Africa-based fintech platform supporting multiple currencies, including cryptocurrency. Implemented the billing service and contributed to a plugin-based architecture that enables additional services and commerce-oriented functionality to be integrated with the platform.

### LEGISLIB

May 2020 – Jul 2020

Legal-document processing and search platform for case-law materials from Bangladesh, India, the UK, and the USA. Processed and reformatted PDF documents and indexed legal content to provide searchable access for Bangladeshi legal professionals and students.

## ACADEMIC & PROGRAMMING ACHIEVEMENTS

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|------|---------------------------------------------------------------|
| 2018 | Participant, ACM ICPC Dhaka Preliminary Contest               |
| 2018 | <b>4th place</b> , DIU Intra Team Programming Contest         |
| 2017 | <b>2nd place</b> , DIU Intra Team Programming Contest         |
| 2017 | <b>9th place</b> , DIU Takeoff Individual Programming Contest |

## CERTIFICATIONS & COURSES

|          |                                                                                   |
|----------|-----------------------------------------------------------------------------------|
| Jun 2025 | Advanced Certificate for Management Professionals (ACMP 4.0), University of Dhaka |
| Apr 2023 | AWS Certified Cloud Practitioner                                                  |
| 2021     | Object-Oriented Design, Coursera                                                  |
| 2021     | The Ultimate Docker Course, Mosh Hamedani                                         |

## TEACHING AND MENTORING

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| Dec 2018<br>Jan 2018 | <b>ACM Co-Chair   Computer &amp; Programming Club, Daffodil International University</b> <ul style="list-style-type: none"> <li>➤ Coordinated programming contests and other ACM-related club activities</li> <li>➤ Mentored students in competitive programming and encouraged participation in programming contests</li> </ul>                                                                               |
| Jan 2019<br>May 2017 | <b>Student Prefect   Daffodil International University</b> <ul style="list-style-type: none"> <li>➤ Assisted students with Competitive Programming and Algorithms through practical lab sessions and academic support</li> <li>➤ Assisted the course instructor with laboratory examinations and student performance evaluation</li> <li>➤ Mentored students requiring additional academic guidance</li> </ul> |

## LANGUAGES

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- › **Bengali** — Native
- › **English** — B2 (CEFR), IELTS Academic 6.5
- › **Hindi** — Conversational